Class 1: Difference-in-Differences

In-class student worksheet

Name: _____ Date: _____

Learning goals: construct a counterfactual, calculate and interpret DiD, and explain the assumption needed for a causal interpretation.

1. Why is a simple comparison insufficient?

Suppose State A has higher average UI benefits and longer average UI receipt than State B.

Can we conclude that higher benefits caused longer UI receipt? Give **two other explanations** for the difference between the states.

2. A policy change: New Jersey and Pennsylvania

In 1996, New Jersey temporarily made 13 additional weeks of UI benefits available. Pennsylvania did not introduce the same extension. We compare regular-benefit exhaustion rates before and after the extension.

Outcome: the percentage of claimants who use up their *regular* UI benefits. It is not the unemployment rate and does not measure exhaustion of the additional extension.

A. Identify the design.

Treated state: _____ Comparison state: _____

What is the treatment? _____

B. Complete the table. All differences are in **percentage points**.

	Before	After	After − Before
New Jersey	49.9%	49.4%	
Pennsylvania	31.8%	29.0%	
NJ − PA			

C. Calculate the DiD estimate using both approaches.

$$\widehat{\tau} = (\text{NJ after} - \text{NJ before}) - (\text{PA after} - \text{PA before}) = _____$$

$$\widehat{\tau} = (\text{NJ} - \text{PA})_{after} - (\text{NJ} - \text{PA})_{before} = _____$$

3. Construct the missing counterfactual

A. Assume that, without the extension, NJ's exhaustion rate would have changed by the same amount as PA's. Complete:

$$\text{NJ after without extension} = \underbrace{_____}_{\text{NJ before}} + \underbrace{_____}_{\text{PA's change}} = _____\%.$$

B. On the axes below, plot both states' before/after outcomes. Add NJ's counterfactual after-period point. Label the estimated policy effect.



C. Interpret your estimate in one sentence. Explain why the estimated effect can be positive even though NJ's exhaustion rate fell.

4. When is the comparison credible?

A. Parallel trends. Complete the statement in words:

Without the extension, New Jersey's exhaustion rate would have...

B. Does DiD require NJ and PA to have the same exhaustion rate before the policy? Explain briefly.

C. Suppose a major employer in NJ closed at the same time as the extension, while no similar event occurred in PA. Why would this threaten our causal interpretation?

Regression connection. In the equation below, circle the coefficient that corresponds to your DiD estimate:

$$Y_{i,t} = \alpha + \beta NJ_i + \lambda After_t + \tau(NJ_i \times After_t) + \varepsilon_{i,t}.$$

$NJ_i = 1$ for NJ and 0 for PA; $After_t = 1$ after the extension and 0 before. This is the simplified classroom comparison based on the Card and Levine (2000) example, not a reproduction of the paper's full regression analysis.